Module 1: Setup & Installation

Deploying Keycloak and understanding its core architecture

From Zero to Running

Laying the foundation for our identity management server.

Module Objectives

1. Docker Setup

Learn how to deploy and run a fully functional Keycloak server locally in seconds using Docker containers.

2. UI Walkthrough

Navigate the Keycloak Admin Console. Discover where everything lives and how to manage your server visually.

3. Core Concepts

Understand the foundational building blocks of Keycloak: Realms, Users, and Clients.

Running with Docker

Deploying identity servers used to require complex Java setups, database configurations, and environment tweaking.

We are going to skip the headache. Using **Docker**, we can pull the official Keycloak image and spin up a pristine, isolated server in a single command.

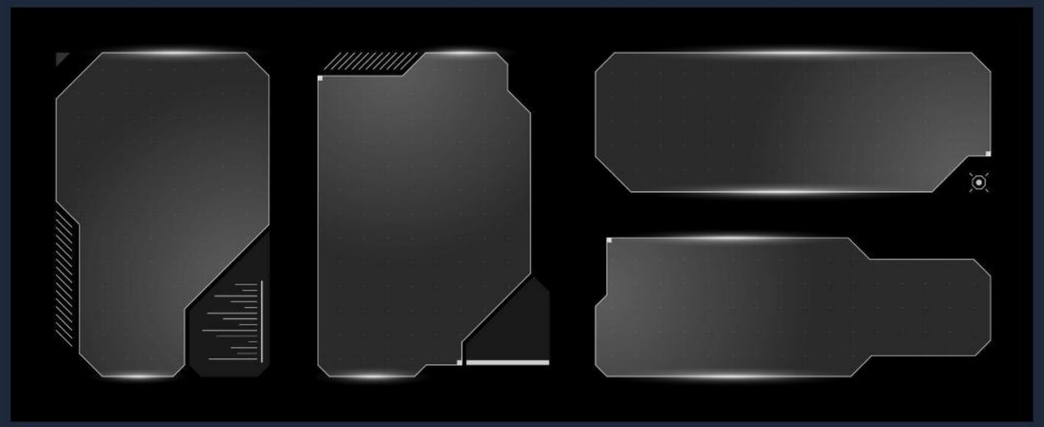
This ensures our development environment is completely consistent, reproducible, and ready for experimentation.

Admin Console Walkthrough

Once Keycloak is running, your primary tool will be the **Admin Console**. This is your central command center.

Through this interface, you will learn how to configure server settings, manage user sessions, set up identity brokering, and tweak authentication flows.

The beauty of Keycloak is that complex security policies can be enforced entirely through this UI, without writing a single line of backend code.



The Core Concepts

Realms

A completely isolated tenant or space. Users, clients, and roles in one realm are completely separate from another. You will create custom realms for your apps.

Clients

The actual applications—whether they are React SPAs, mobile apps, or backend Spring Boot APIs—that request authentication from your Keycloak server.

Users

The actual individuals, entities, or service accounts that log into your system to consume resources. Managed securely within their respective realm.

Module Roadmap

1. Spin Up

Execute Docker commands to start the Keycloak container locally.

2. Access UI

Log into the master realm and explore the Admin dashboard.

3. Create Realm

Set up an isolated tenant space specifically for our course project.

4. Base Config

Define our first basic Client and test User within the new realm.

Let's Get Hands-On

Open your terminal and let's deploy Keycloak.

Image Sources

https://png.pngtree.com/thumb_back/fw800/background/20251008/pngtree-abstract-network-of-glowing-blue-nodes-and-lines-on-dark-background-image_19804182.webp

Source: pngtree.com

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https://static.vecteezy.com/system/resources/previews/067/928/255/non_2x/high-tech-futuristic-hud-frame-set-with-abstract-sci-fi-dashboard-glowing-lines-and-cyber-display-vr-interface-mixes-modern-data-layout-with-holographic-control-visuals-vector.jpg

Source: www.vecteezy.com